

## Preliminary work.

This book assumes that you are already familiar with a number of mathematical ideas from your mathematical studies in earlier years.

This section outlines the ideas which are of particular relevance to Unit Three of the Mathematics Applications course and for which some familiarity will be assumed, or for which the brief explanation given here may be sufficient to bring your understanding of the concept up to the necessary level.

Read this "preliminary work" section and if anything is not familiar to you, and you don't understand the brief explanation given here, you may need to do some further reading to bring your understanding of those concepts up to an appropriate level for this unit. (If you do understand the work but feel somewhat "rusty" with regards to applying the ideas some of the chapters afford further opportunities for revision as do some of the questions in the miscellaneous exercises at the end of chapters.)

- ☞ Chapters in this book will continue some of the topics from this preliminary work by building on the assumed familiarity with the work.
- ☞ The miscellaneous exercises that feature at the end of each chapter may include questions requiring an understanding of the topics briefly explained here.

- **Number.**

The understanding and appropriate use of the rule of order, fractions, decimals, percentages, rounding, truncation, square roots and cube roots, numbers expressed with positive integer powers, e.g.  $2^3$ ,  $5^2$ ,  $2^5$  etc, expressing numbers in standard form, e.g.  $2.3 \times 10^4$  ( $= 23\,000$ ),  $5.43 \times 10^{-7}$  ( $= 0.000\,000\,543$ ), also called scientific notation, and familiarity with the symbols  $>$ ,  $\geq$ ,  $<$ , and  $\leq$  is assumed.

- **Percentages.**

It is assumed that you are able to express one quantity as a percentage of another quantity.

E.g. 21 out of 50 is 42%. 
$$\frac{21}{50} \times 100$$

35 out of 78 is 44.87%, correct to two decimal places. 
$$\frac{35}{78} \times 100$$

197 out of 5063 is 4%, to the nearest whole percent. 
$$\frac{197}{5063} \times 100$$

It is also assumed that you are able to increase or decrease an amount by a certain percentage by appropriate multiplication.

For example: To increase an amount by 25% we multiply by 1.25.

To decrease an amount by 5% we multiply by 0.95.

- **Data.**

Data collection is often carried out to investigate some aspect of our lives. The methods by which we collect that data can vary as can the types of data we collect. For some investigations we collect the data ourselves by asking questions, by measuring, by experiment etc. This is called **primary data** - data we have collected ourselves.

Sometimes it is appropriate to use the data already collected by others. For us this would be **secondary data** - data collected by others.

If we ask someone what their favourite colour is or how tall they are the answers we get will vary from one person to another. Not everyone has blue as their favourite colour, not everyone is 163 cm tall etc. The responses will vary. Favourite colour and height are examples of **variables**.

If we are considering one variable, for example favourite colour, any data we collect will be **univariate**. However if we wanted to investigate how favourite colour may change with the age of a person we would be considering two variables – favourite colour and age. In such cases we would be considering **bivariate** data.

Consider the following questions that might be asked in some data collection activities:

How do you get to school (or work) – walk, cycle, car, bus, train, other?

Are you male or female?

What is your favourite colour?

How would you rank your fitness level – low, medium, high?

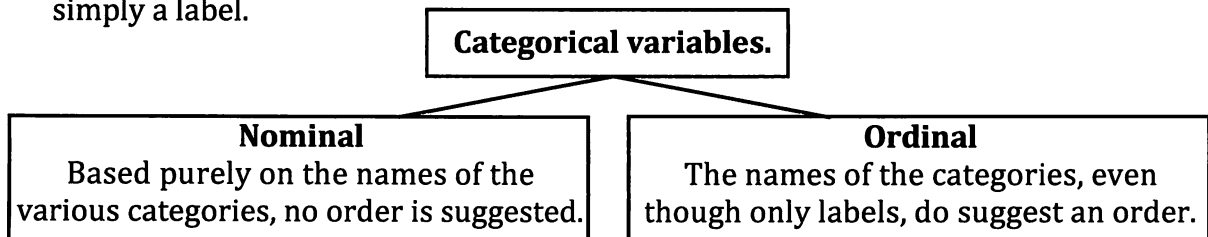
What is the number of the house you live in?

Each of these questions allow us to place the person who responds into a particular **category** or **group**. For example the group of people who walk to school, or the group of males, or the group of people whose favourite colour is red, or the group of people who consider their fitness level to be high, or the group of people who live in even numbered houses, etc. The variable concerned, be it mode of transport, gender, favourite colour, fitness level or house number are all examples of **categorical variables**. Data associated with a categorical variable is called **categorical data**.

Notice that the first three questions each involve categories for which order is irrelevant. In general it would be pointless to suggest an order to the modes of transport, or to the gender categories, or place red ahead of blue as a category. We may rank order according to the numbers of people in that category but not on the basis of the category *names* themselves which have no natural order about them. Such categorical variables are called **nominal categorical variables**. (Name – nominal).

However the last two question involve categories that do have a natural *order* about them. This question involves an **ordinal categorical variables**, i.e. a categorical variable that does have some natural order about it. (Order – ordinal)

Notice though that whilst categorical variables can have categories that have numbers assigned to them these numbers are simply labels, they have no numerical significance. Houses numbered 11, for example, are not necessarily bigger, better or more expensive than houses numbered 10. The house number is simply a label.



Now consider the following questions that might be asked in some data collection activities:

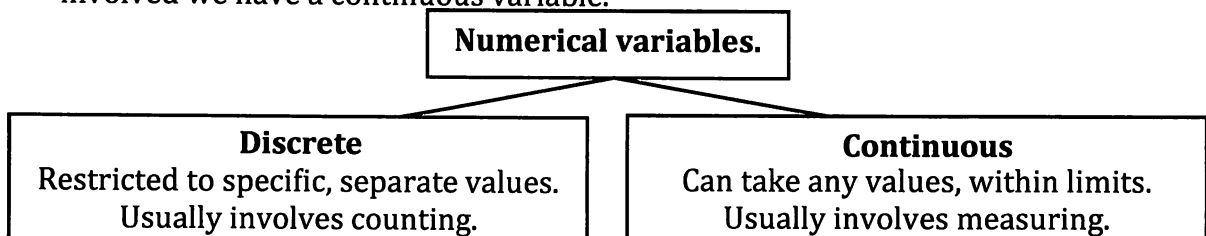
- How many people live in your house?
- How many people in your Mathematics class?
- How tall are you?
- How far have you walked today?

The responses to each of these questions will be **numerical** and the number will not just be a label, it will indicate a size or an amount. Some form of counting or measuring will be required to be done, or to have been done, for the response to be given. In each case the variable concerned, be it the number of people living in your house, the number of people in your Mathematics class, how tall you are, or how far you have walked today, are all examples of **numerical variables**. Data associated with a numerical variable is called **numerical data**.

Notice that the first two questions each involve responses that can only take specific values, in this case integer values. We cannot have 2.3 people living in a house, we cannot have 28.6 people in a Mathematics class. Numerical variables that take specific values taken from a finite set of values, usually the counting numbers 0, 1, 2, 3, ..., are called **discrete variables**. Data associated with discrete variables is **discrete data**.

However the last two questions each involve responses that can take any value (usually within some realistic range). We can be 164.5 cm tall, we can walk 5.278 km in a day. Responses no longer have to be integer values and can now be taken from an infinite set of possible values. In practice the limit on the values taken are those of reasonableness (eg we cannot have 18.24 metres for the height of someone) and the accuracy of the measurement instrument used. Numerical variables that can take any value in an interval are called **continuous variables**. Data associated with a continuous variable is **continuous data**.

Generally, if counting is involved we have a discrete variable, if measurement is involved we have a continuous variable.



- **Measures of central tendency.**

It is assumed that you are familiar with using the **mean**, the **median** and the **mode** as measures that can be used to summarise a set of data.

The mean and the median each indicate a "central score". The mode is often included in these "averages" but there is no guarantee that the mode is any "central" measure.

- **Measures of dispersion.**

It is assumed that you are familiar with using the **range** and the **standard deviation** to describe the *spread* of data.

The range of a set of scores is the difference between the highest score and the lowest score and gives a simple measure of how widely the scores are spread. Whilst the range is easy to calculate it is determined using just two of the scores. For this reason it is of limited use.

The standard deviation is more representative of spread because it is obtained using all of the scores in a set, not just the highest and lowest

The standard deviation is found by finding how much each of the scores differs from the mean, squaring these values, finding the average of the squared values and then taking the square root of this average.

Many calculators can determine the standard deviation of a set of scores.

- **The statistical investigation process.**

Collecting data is one part of the *statistical investigation process*, a process that can be described by the following steps:

- ① Clarify the problem and formulate one or more questions that can be answered with data.
- ② Design and implement a plan to collect and obtain appropriate data.
- ③ Select and apply appropriate graphical or numerical techniques to analyse the data.
- ④ Interpret the results of this analysis and relate the interpretation to the original question and communicate findings in a systematic and concise manner.

(As stated in the Australian Curriculum Glossary for General Mathematics and the West Australian Glossary for Mathematics Applications.)

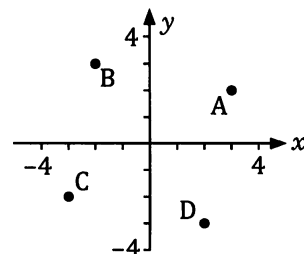
- **Graphical display of data.**

It is assumed that you are familiar with using box plots and histograms to compare data sets.

- **Coordinates.**

It is assumed that you are familiar with the idea that points on a graph can be located by stating the coordinates of the point.

For example point A has coordinates (3, 2),  
 point B has coordinates (-2, 3),  
 point C has coordinates (-3, -2),  
 point D has coordinates (2, -3).



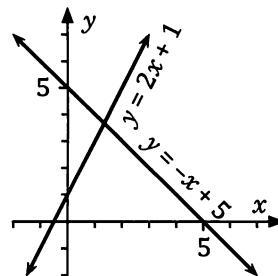
- **Straight line graphs.**

If points lie in a straight line their coordinates,  $(x, y)$ , obey a rule of the form  $y = mx + c$  and the relationship between  $x$  and  $y$  is said to be *linear*. In this rule  $m$  is the gradient of the straight line and the point  $(0, c)$  is where the line cuts the vertical axis.

Hence the line  $y = 2x + 1$  has a gradient of 2 and cuts the  $y$ -axis at  $(0, 1)$ .

The line  $y = -x + 5$  has gradient of  $-1$  and cuts the  $y$ -axis at  $(0, 5)$ .

In the equation  $y = mx + c$  the value of  $m$ , the gradient, tells us the amount by which  $y$  increases for each unit increase in  $x$ .



- **Solving equations.**

It is assumed that you are already familiar with the idea that solving an equation involves finding the value(s) the unknown can take that make the equation true.

For example,  $x = 5.5$  is the solution to the equation  $15 - 2x = 4$   
because  $15 - 2(5.5) = 4$ .

To solve an equation, eg  $15 - 2x = 4$ , we could proceed *mentally*:

We know that fifteen take *eleven* equals four.

Thus  $2x = 11$  and so  $x = 5.5$ .

We could use the *solve facility* of some calculators.

Or we could use a *step by step approach* to isolate  $x$ :

|                                                       |                      |
|-------------------------------------------------------|----------------------|
| Given the equation:                                   | $15 - 2x = 4$        |
| Add $2x$ to both sides to make the $x$ term positive: | $15 = 4 + 2x$        |
| Subtract 4 from both sides to isolate $2x$ :          | $15 - 4 = 2x$        |
|                                                       | $\therefore 11 = 2x$ |
| Divide both sides by 2 to isolate $x$ :               | $5.5 = x$            |
|                                                       | $\therefore x = 5.5$ |

It is also anticipated that you are able to solve more involved linear equations, for example:

$$5x - 7 - 2x - 8 = 30 - 6x \qquad \text{Solution: } x = 5$$

$$\frac{x+3}{2} - 1 = 5 \qquad \text{Solution: } x = 9$$

$$2(x+3) - 3(2x+1) = -5 \qquad \text{Solution: } x = 2$$

and simultaneous linear equations, for example:

|                                                        |                                                         |                                                         |
|--------------------------------------------------------|---------------------------------------------------------|---------------------------------------------------------|
| $\begin{cases} 3x + 2y = 11 \\ x + 2y = 1 \end{cases}$ | $\begin{cases} 5x - 2y = 6 \\ 3x + 2y = 26 \end{cases}$ | $\begin{cases} 2x + 3y = 12 \\ x + 4y = 11 \end{cases}$ |
| Solutions: $x = 5, y = -2$                             | $x = 4, y = 7$                                          | $x = 3, y = 2$                                          |

- **Matrices.**

It is assumed that you are familiar with the rows and columns, *rectangular array*, form of presenting information called a **matrix**, an example of which is shown on the right.

This matrix has 6 rows and 6 columns. We say it is a *six by six* matrix, (written  $6 \times 6$ ). This notation gives the **size** or **dimensions** of the matrix.

Matrices do not have to have the same number of rows as they have columns, however those that do are called **square matrices**.

The matrix on the right is a  $6 \times 6$  square matrix.

Some matrices of other sizes are shown below.

$$\begin{bmatrix} 1 & 0 & 4 \\ 3 & 2 & 0.5 \end{bmatrix}$$

2 rows  
and  
3 columns.

A  $2 \times 3$  matrix.

$$\begin{bmatrix} 2 & 5 \\ 11 & -2 \end{bmatrix}$$

2 rows  
and  
2 columns.

A  $2 \times 2$  matrix.

(A square matrix.)

$$\begin{bmatrix} 3 \\ 2 \\ -1 \end{bmatrix}$$

3 rows  
and  
1 column.

A  $3 \times 1$  matrix.

$$\begin{bmatrix} 1 & 0 & 2 & 3 \\ 1 & 2 & -1 & 0 \\ 0 & 1 & 0 & 3 \end{bmatrix}$$

3 rows  
and  
4 columns.

A  $3 \times 4$  matrix.

A matrix consisting of just one column, like the third matrix above, is called a **column matrix**.

Any matrix consisting of just one row is called a **row matrix**. For example:

$$\begin{bmatrix} 5 & 0 & -2 & 1 \end{bmatrix}$$

A square matrix having zeros in all spaces that are not on the **leading diagonal** is called a **diagonal matrix**.

$$\begin{bmatrix} \text{leading diagonal} \end{bmatrix}$$

$$\begin{bmatrix} 5 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & -4 \end{bmatrix}$$

We commonly use capital letters to label different matrices. The corresponding lower case letters, with subscripted numbers, are then used to indicate the row and column a particular entry or **element** occupies.

For the matrix A shown on the right the element occupying the 3rd row and 2nd column is the number 7. Thus  $a_{32} = 7$ .

Similarly

$$a_{11} = 2,$$

$$a_{12} = 0,$$

$$a_{13} = -1, \text{ etc.}$$

$$A = \begin{bmatrix} 2 & 0 & -1 & 3 \\ 1 & 6 & -3 & 9 \\ 5 & 7 & 8 & 4 \end{bmatrix}$$

**To add (or subtract) matrices** we add (or subtract) the elements occurring in corresponding locations. To do this there must be elements in corresponding locations. Thus we can only add or subtract matrices that are the same size as each other. For example:

$$\text{if } A = \begin{bmatrix} 1 & 0 & 2 & 3 \\ 4 & -2 & 3 & 5 \\ 2 & 1 & -3 & 4 \end{bmatrix} \text{ and } B = \begin{bmatrix} 2 & 1 & -3 & 2 \\ 5 & 1 & 2 & 4 \\ 3 & 2 & 0 & -5 \end{bmatrix} \text{ then } A + B = \begin{bmatrix} 3 & 1 & -1 & 5 \\ 9 & -1 & 5 & 9 \\ 5 & 3 & -3 & -1 \end{bmatrix}$$

$$\text{and similarly } A - B = \begin{bmatrix} -1 & -1 & 5 & 1 \\ -1 & -3 & 1 & 1 \\ -1 & -1 & -3 & 9 \end{bmatrix}$$

**To multiply a matrix by a number** we multiply each element of the matrix by that number. (This is referred to as "multiplication by a **scalar**".)

**Equal matrices** must be the same size and have all corresponding elements equal.

$$\text{Thus if } \begin{bmatrix} a & b & c \\ d & e & f \end{bmatrix} = \begin{bmatrix} 2 & 3 & -5 \\ 1 & 0 & -2 \end{bmatrix} \text{ then } \begin{matrix} a=2 & b=3 & c=-5 \\ d=1 & e=0 & f=-2 \end{matrix}$$

**Matrices can be multiplied together if the number of columns in the first matrix equals the number of rows in the second matrix.**

To multiply two matrices we "spin" each row of the first matrix to align with each column of the second, multiply and then add:

$$\text{If } A = \begin{bmatrix} 2 & 1 & 3 \\ 0 & -1 & 2 \end{bmatrix} \text{ and } B = \begin{bmatrix} 1 & 2 \\ 4 & -1 \\ 1 & -3 \end{bmatrix} \text{ the product } AB \text{ is found as shown below.}$$

$$\begin{bmatrix} 2 & 1 & 3 \\ 0 & -1 & 2 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 4 & -1 \\ 1 & -3 \end{bmatrix} = \begin{bmatrix} (2)(1) + (1)(4) + (3)(1) & (2)(2) + (1)(-1) + (3)(-3) \\ (0)(1) + (-1)(4) + (2)(1) & (0)(2) + (-1)(-1) + (2)(-3) \end{bmatrix}$$

$$= \begin{bmatrix} 9 & -6 \\ -2 & -5 \end{bmatrix}$$

Many calculators will accept data in matrix form and can then manipulate these matrices in various ways.

$$\begin{bmatrix} 1 & 2 & 4 \\ 0 & -4 & 5 \end{bmatrix} + \begin{bmatrix} 3 & 5 & -2 \\ 1 & 0 & -2 \end{bmatrix}$$

$$\begin{bmatrix} 4 & 7 & 2 \\ 1 & -4 & 3 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 1 & 3 \\ 0 & -1 & 2 \end{bmatrix} \times \begin{bmatrix} 1 & 2 \\ 4 & -1 \\ 1 & -3 \end{bmatrix}$$

$$\begin{bmatrix} 9 & -6 \\ -2 & -5 \end{bmatrix}$$

Note: In the product  $AB$  we say that  $B$  is *premultiplied* by  $A$   
or that  $A$  is *postmultiplied* by  $B$ .

• **Networks and matrices.**

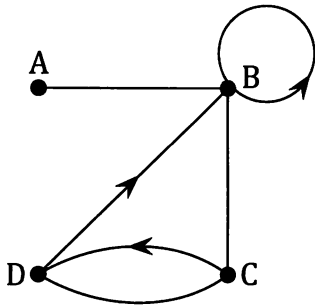
If you have previously completed *Unit One of Mathematics Applications* you will be familiar with

- (a) the idea of road systems and social interactions being displayed in the diagrammatic form shown below (called networks),  
and (b) the associated matrix for such systems.

For example:

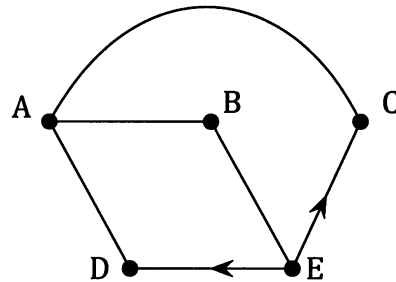
A road network.

The network shows road links between towns A, B, C and D.



A social interaction network.

The network shows who in a group of five people, A, B, C, D and E has visited the house of somebody else in the group.



The associated matrices:

|      |   | To |   |   |   |                                                                                                                    |
|------|---|----|---|---|---|--------------------------------------------------------------------------------------------------------------------|
|      |   | A  | B | C | D |                                                                                                                    |
| From | A | 0  | 1 | 0 | 0 | $\left[ \begin{array}{ccccc} 0 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 \\ 0 & 1 & 0 & 2 \\ 0 & 1 & 1 & 0 \end{array} \right]$ |
|      | B | 1  | 1 | 1 | 0 |                                                                                                                    |
|      | C | 0  | 1 | 0 | 2 |                                                                                                                    |
|      | D | 0  | 1 | 1 | 0 |                                                                                                                    |
|      | E | 0  | 1 | 1 | 0 |                                                                                                                    |

|   |   | A | B | C | D | E |
|---|---|---|---|---|---|---|
| A | A | 0 | 1 | 1 | 1 | 0 |
|   | B | 1 | 0 | 0 | 0 | 1 |
|   | C | 1 | 0 | 0 | 0 | 0 |
|   | D | 1 | 0 | 0 | 0 | 0 |
|   | E | 0 | 1 | 1 | 1 | 0 |

Notice that in the social interaction matrix:

- We use a "1" to indicate "has visited" and a "0" to indicate "has not visited".
- We use the convention that the matrix shows "what the row person does to the column person". For example the entry in row 5 (person E) and column 4 (person D) shows that person E has been to the house of person D.
- The "has visited your own house" is really rather pointless information so we simply place zeros on the leading diagonal. On some occasions like this we might simply decide to put dashes on the leading diagonal. (However if we later wanted to multiply the matrix by a scalar or another matrix these dashes could cause a problem.)